2.5 Limits at infinity and horizontal asymptotes.

<u>Definition 1</u>: Let f be a function defined on some interval $(a, +\infty)$. Then

$$\lim_{x \to +\infty} f(x) = L$$

means that the values $f(x)$ can be made arbitrarily close to L by taking x sufficiently large.

<u>Definition 2</u>: Let f be a function defined on some interval $(-\infty, a)$. Then

$$\lim_{x \to -\infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently large negative.

<u>Example 1</u>: Determine: (a) $\lim\limits_{x \to -\infty} e^{x}$ (b) $\lim\limits_{x \to +\infty} \dfrac{x^2-1}{x^2+1}$

Definition 3: The line $y=L$ is called a horizontal asymptote of the curve $y=f(x)$ if either

$$\lim_{x\to+\infty} f(x)=L \quad \text{or} \quad \lim_{x\to-\infty} f(x)=L$$

Example 2: Find the horizontal asymptotes of the functions of Example 1.

$$\boxed{\lim_{x\to-\infty} \tan^{-1}(x) = -\frac{\pi}{2} \quad , \quad \lim_{x\to+\infty} \tan^{-1}(x) = \frac{\pi}{2}}$$

Theorem 1: If $r>0$ is a rational number, then

$$\lim_{x\to+\infty} \frac{1}{x^r} = 0$$

If $r>0$ is a rational number such that x^r is defined for all x, then

$$\lim_{x\to-\infty} \frac{1}{x^r} = 0$$

Example 3: Find the horizontal asymptotes, if any of

$$g(x) = \frac{4x^4 - 3x^2 + 2x}{7x^3 + x^2}$$

Example 4 : Find $\lim\limits_{x\to+\infty}\left(\sqrt{x^2+1}-x\right)$

Definition 4 :

- $\lim\limits_{x\to+\infty} f(x) = +\infty$ means that the values of $f(x)$ become large as x becomes large
- $\lim\limits_{x\to+\infty} f(x) = -\infty$ means that the values of $f(x)$ become large negative as x becomes large.
- $\lim\limits_{x\to-\infty} f(x) = +\infty$ means that the values of $f(x)$ become large as x becomes large negative
- $\lim\limits_{x\to-\infty} f(x) = -\infty$ means that the values of $f(x)$ become large negative as x becomes large negative

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Example 5 : Find (a) $\lim\limits_{x\to-\infty}(x^2+x)$

(b) $\lim\limits_{x\to+\infty}\dfrac{3x^2+7\sqrt{x}}{2x^3-x}$

If $f(x)$ is a rational function, i.e $f(x) = \frac{P(x)}{Q(x)}$

$$\deg(P) = \deg(Q) \overset{\text{We expect}}{\Longrightarrow} \lim_{x \to +\infty} f(x) = L_1,\ L_1 \in \mathbb{R}$$

and $\lim_{x \to -\infty} f(x) = L_2,\ L_2 \in \mathbb{R}$

$$\deg(P) < \deg(Q) \overset{\text{We expect}}{\Longrightarrow} \lim_{x \to +\infty} f(x) = \lim_{x \to -\infty} f(x) = 0$$

$$\deg(P) > \deg(Q) \overset{\text{We expect}}{\Longrightarrow} \lim_{x \to +\infty} f(x) = +\infty \text{ or } -\infty$$

and

$$\lim_{x \to -\infty} f(x) = +\infty \text{ or } -\infty$$

YOU MUST SHOW YOUR CALCULATION

$$\frac{\infty}{\infty}\ ;\ \frac{0}{0}\ ;\ \infty - \infty\ ;\ 0 \times \infty$$

are all indeterminate forms

Example 6: Find the horizontal asymptotes of

$$f(x) = \frac{\sqrt{4x^2 - 9}}{3 + 5x}$$

Example 1 :

(a) $\lim\limits_{x\to-\infty} e^{x} = \lim\limits_{x\to-\infty} \frac{1}{e^{x}} = 0$

(b) $\lim\limits_{x\to+\infty} \frac{x^2-1}{x^2+1} = \lim\limits_{x\to+\infty} \frac{x^2\left(1-\frac{1}{x^2}\right)}{x^2\left(1+\frac{1}{x^2}\right)} = \lim\limits_{x\to+\infty} \frac{1-\frac{1}{x^2}}{1+\frac{1}{x^2}}$

$= 1$

since $\lim\limits_{x\to+\infty} \frac{1}{x^2} = 0$

Example 2 :

(a) $y=0$ is the only horizontal asymptote

(b) $y=1$ is a horizontal asymptote

Example 3 :

$$g(x) = \frac{x^4\left(4-\frac{3}{x^2}+\frac{2}{x^3}\right)}{x^3\left(7+\frac{1}{x}\right)}$$

$$\lim_{x\to+\infty} g(x) = \lim_{x\to+\infty} \frac{x\left(4-\frac{3}{x^2}+\frac{2}{x^3}\right)}{7+\frac{1}{x}} = +\infty$$

$$\lim_{x\to-\infty} g(x) = \lim_{x\to-\infty} \frac{x\left(4-\cancelto{0}{\frac{3}{x^2}}+\cancelto{0}{\frac{2}{x^3}}\right)}{7+\cancelto{0}{\frac{1}{x}}} = -\infty$$

No "H.A"

Example 4 :

$$\lim_{x \to +\infty} \left(\sqrt{x^2+1} - x \right) \qquad (= \infty - \infty)$$

Indeterminate form !!

$$\infty - \infty \neq 0$$

let rationalize

$$\lim_{x \to +\infty} \sqrt{x^2+1} - x = \lim_{x \to +\infty} \frac{\left(\sqrt{x^2+1} - x\right)\left(\sqrt{x^2+1} + x\right)}{\sqrt{x^2+1} + x}$$

$$= \lim_{x \to +\infty} \frac{(x^2+1) - (x^2)}{\sqrt{x^2+1} + x}$$

$$= \lim_{x \to +\infty} \frac{1}{\sqrt{x^2+1} + x} = 0$$

Examples 5

(a) $\lim_{x \to -\infty} (x^2 + x) \qquad (= \infty - \infty)$

$$= \lim_{x \to -\infty} x(x+1) = \infty \times (-\infty) = -\infty$$

Note that $\lim_{x \to -\infty} (x^2 - x) = \infty - (-\infty) = \infty + \infty = \infty$

No indeterminate form

(b) $\lim_{x \to +\infty} \dfrac{x^2\left(3+\frac{7}{x\sqrt{x}}\right)}{x^3\left(2-\frac{1}{x^2}\right)} = \lim_{x \to +\infty} \dfrac{3+\cancel{\frac{7}{x\sqrt{x}}}^{\to 0}}{x\left(2-\cancel{\frac{1}{x^2}}^{\to 0}\right)}$

$= 0$

Example 6 : $f(x) = \dfrac{\sqrt{x^2\left(4-\frac{9}{x^2}\right)}}{3+5x} = \dfrac{|x|\sqrt{4-\frac{9}{x^2}}}{3+5x}$

$f(x) = \dfrac{|x|}{x} \cdot \dfrac{\sqrt{4-\frac{9}{x^2}}}{\frac{3}{x}+5}$

$\lim_{x \to +\infty} f(x) = \lim_{x \to +\infty} 1 \times \dfrac{\sqrt{4-\cancel{\frac{9}{x^2}}^{\to 0}}}{\cancel{\frac{3}{x}}^{\to 0}+5} = \dfrac{\sqrt{4}}{5} = \dfrac{2}{5}$

since $x > 0$ and $\dfrac{|x|}{x} = 1$. For $x < 0$, $\dfrac{|x|}{x} = -1$

so

$\lim_{x \to -\infty} f(x) = \lim_{x \to -\infty} -1 \times \dfrac{\sqrt{4-\cancel{\frac{9}{x^2}}^{\to 0}}}{\cancel{\frac{3}{x}}^{\to 0}+5} = -\dfrac{\sqrt{4}}{5} = -\dfrac{2}{5}$

Hence $y = \dfrac{2}{5}$ and $y = -\dfrac{2}{5}$ are H.A .